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Asexual Reproduction

produces offspring
genetically identical to the
parent

- some plants
 - bacteria
- coral, sponges, others
- some animals in weird situations

Sexual Reproduction

genetic information from
two parents combine to
make genetically new
offspring

- cattle
- dogs
- cats
- pea plants

Trait: a characteristic that can be passed on genetically from parent to offspring (i.e. eye color, but not sense of humor)

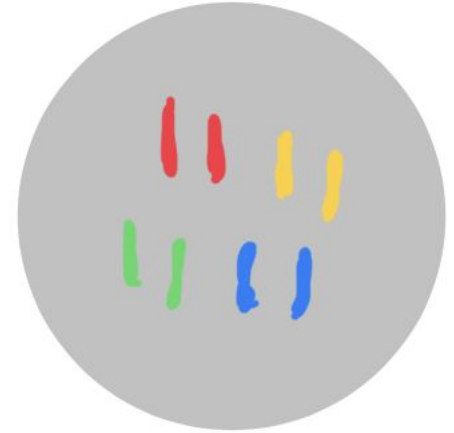
Gene: a bit of DNA that determines a trait (i.e. the genes for eye color or hair texture)

Genes are instructions for cells.

- the eye color gene has instructions for making the brown pigment in eyes, leading to brown eyes
- you have a gene that instructs your feet to have five toes each

offspring get all of their genes from their parents

- organisms have 2 sets of each gene (1 from each parent)

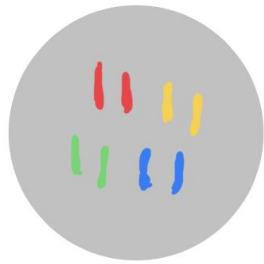


Our example cell has four genes, and there are two copies of each gene.

L1 - Patterns of Reproduction

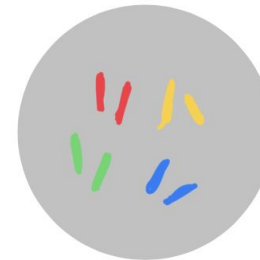
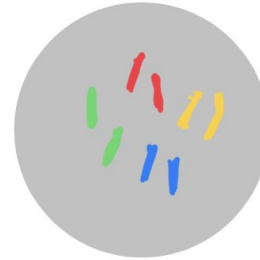
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Mitosis creates genetic copies of the parent cell.



parent cell

daughter cells

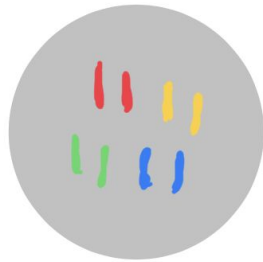


this is how cells replicate in your body! all of your cells have the same DNA.

L1 - Patterns of Reproduction

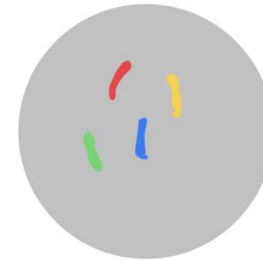
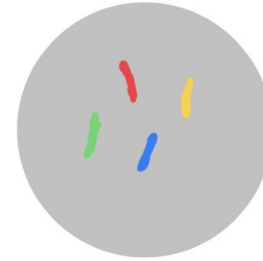
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Meiosis creates cells with half the genetic info of the parent cell (only one copy of each gene)



parent cell

daughter cells



the cells made by **meiosis** will become the egg and sperm cells for sexual reproduction

a different type of the same gene is an allele

The “Having Arms” gene:

allele 1 (A): a complete set of instructions. results in Having Arms.

allele 2 (a): a janky (mutated) “Having Arms” gene, no instructions for how to have arms. results in No Arms.



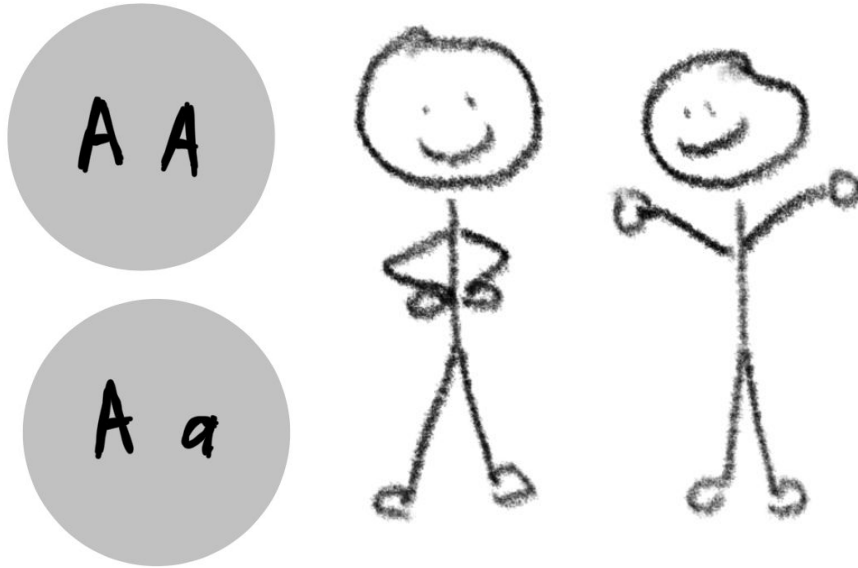
A



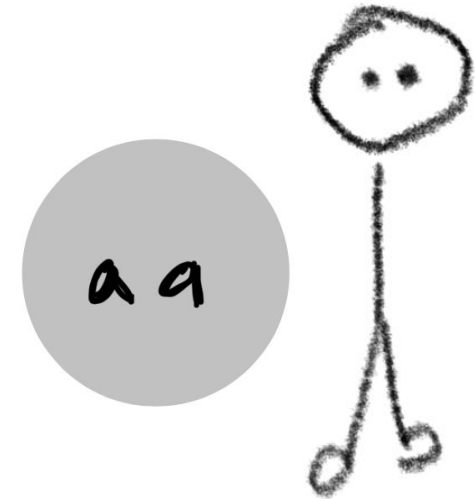
a

L1 - Patterns of Reproduction

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Individuals with at least one working allele (A) have arms.



Individuals with no working alleles, only janky alleles (a) have no arms.

Sometimes alleles can be dominant or recessive.

- dominant alleles are always represented (A - having arms)
- recessive alleles are “hidden” behind dominant alleles (a - no arms)
 - u only *don't* have arms if you only have copies of the janky allele

dominant $\neq$ most common

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L2 Patterns of Inheritance



Genotype: the combination of alleles for a gene

Phenotype: the resulting expression of the gene

Ex: A genotype of Aa will have a phenotype of Having Arms, whereas a genotype of aa will have a phenotype of No Arms.

Other patterns of Inheritance

- genetics is ridiculously complicated
- most traits are influenced by multiple genes (height, metabolism, skin color)

WHY LEARN THIS STUFF??

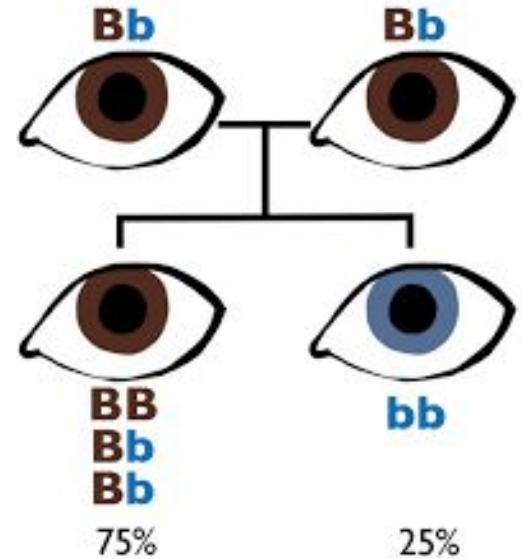
1. Inheritance of genetic material is passed down from parents to offspring randomly.
2. Sexual reproduction creates new combinations of alleles and therefore new genotypes and phenotypes.

L1 - Patterns of Reproduction

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brown eye color is dominant over blue

two brown-eyed parents can have a blue eyed child



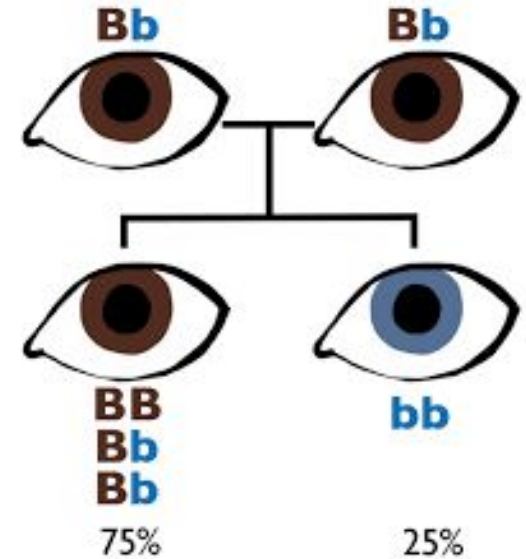


Two genes control coat color in labs. One gene decides on black or brown fur, a second gene controls whether the first gene is even expressed. If a dog has two recessive alleles for the second gene, neither the black nor brown phenotype will be expressed.



Dominant vs Recessive

- dominant alleles are always represented (brown allele)
 - if u have a brown allele, u will have brown eyes
- recessive alleles are “hidden” behind dominant alleles (blue)
 - u need two copies of the blue allele to have blue eyes



Human Activities

Agriculture

- cow farts (methane is the beefiest GG)

Burning Fossil Fuels

- burning oil and coal releases CO₂ and energy

Industry (factories and whatnot)

- GG produced as side products. Cement Bad.

Deforestation

- Trees take in CO₂, so deforestation Bad.

HOW TO UNCHANGE CLIMATE
REDUCE GREENHOUSE GAS
EMISSIONS

HOW TO DO THAT
UHHHHHHHH it's complicated

HOW TO:

- with alternative energy
- energy-efficient tech
- “New solutions”
- THE GOVERNMENT HAS GOTTA STEP IN
otherwise, choices are made based on MONEY.

Dealing with Climate Change

The effects of climate change may look grim, but there are ways humans can alter their behaviors to reduce or counteract the threat of global warming. Scientists, lawmakers, engineers, and other creative minds are working together to find new solutions to reduce the extent of climate change and human vulnerability to its effects. Because these changes have accumulated over time, it will also take time to slow or reverse the effects.

Alternative Energy Human activities require energy, and 91 percent of all greenhouse gases emitted by human activities comes from burning fossil fuels. By developing alternative energy, or energy sources that do not come from fossil fuels, we can reduce the extraction and burning of fossil fuels, thereby reducing greenhouse gas emissions. Alternative energy sources such as solar, wind, and tidal power do not require fossil fuels and have no greenhouse gas emissions.

Alternative Energy Examples:

SOLAR

WIND

NUCLEAR

HYDROELECTRIC

GEOTHERMAL

Absorb

energy is taken in and converted into thermal energy (or chemical energy, if ur a chloroplast)

Reflect

energy “bounces” off and goes somewhere else

different substances reflect and absorb radiation differently.

Materials

2 styrofoam cups

2 thermometers

Hot water

Aluminum foil

**COPY ALL THIS INTO
YOUR NOTEBOOK :)**

Procedure Summary

Experiment 1: Volume

Pour different amounts of water into cups with water of the same temperature and monitor temp for 10 minutes.

Experiment 2: Lid

Pour the same amount of water into two cups, but cover one with aluminum foil. Monitor temp for 10 minutes.